Compiled Solutions – Senior Division

1. 01-02 C1 Recursive Functions

$f(1) = f(3) + 1 = 14 + 1 = 15$ — 15
$f(3) = f(5) + 3 = 11 + 3 = 14$
$f(5) = f(7) + 5 = 6 + 5 = 11$
$f(7) = 7 - \ = 6$

2. 01-02 C1 Recursive Functions

$f(9,4) = f(7,8) + 1 = 1 - 11 = -10$ — –10
$f(7,8) = f(5,16) + 1 = 1 - 12 = -11$
$f(5,16) = f(2,15) + 2 = 2 - 14 = -12$
$f(2,15) = f(-1,14) + 2 = 2 - 16 = -14$
$f(-1,14) = -2 - 14 = -16$

3. 02-03 C1 Recursive Functions

$F(9) = F(F(7) + 1) = F(3) = 2$ — 2
$F(7) = F(F(5) + 1) = F(3) = 2$
$F(5) = F(F(3) + 1) = F(3) = 2$
$F(3) = F(2) - 1 = 3 - 1 = 2$
$F(2) = F(1) - 1 = 4 - 1 = 3$
$F(1) = 1 + 3 = 4$

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4. 02-03 C1 Recursive Functions

The first time through the function gives 1 painted triangle as shown. — 121
The second recursion gives 3, the third gives 9, the fourth gives 27 and the fifth gives 81. $1 + 3 + 9 + 27 + 81 = 121$. This is Sierpinski's Triangle.

5. 03-04 C1 Recursive Functions

$f(4) = f(1) + 3 = (1^2 + 2) + 3 = 3 + 3 = 6$ — 18
$f(6) = f(3) + 3 = 7 + 3 = 10$
$f(10) = f(7) + 3 = (f(4) + 3) + 3 = (6 + 3) + 3 = 12$
$f(12) = f(9) + 3 = (f(6) + 3) + 3 = (10 + 3) + 3 = 16$
$f(16) = f(13) + 3 = (f(10) + 3) + 3 = (12 + 3) + 3 = 18$

6. 03-04 C1 Recursive Functions

$f(5) = f(f(4)) + 5 = f(1) + 5 = -2 + 5 = 3$ — 3
$f(4) = f(f(3)) + 4 = f(0) + 4 = -3 + 4 = 1$
$f(3) = 3 - 3 = 0$ $\quad f(0) = 0 - 3 = -3$
$f(1) = 1 - 3 = -2$

7. 04-05 C1 Recursive Functions

$f(5,1) = f(f(1,4),5) + 1 = f(2,5) + 1 = 2 + 1 = 3$ 3
$f(1,4) = f(3, 3\ ) - 1 = 3 - 1 = 2$
$f(3,3) = 3$
$f(2,5) = f(4,4) - 2 = 4 - 2 = 2$
$f(4,4) = 4$

8. 04-05 C1 Recursive Functions

There are 2 8x8, 4 4x4, 8 2x2, 16 1x1, and 8 ½x½ squares. 248
That gives you $2*64+4*16+8*4+16*1+32*1/4 = 128+64+32+16+8 = 248$.

9. 05-06 C1 Recursive Functions

$F(10) = F(8) - 5 = 3 - 5 = -2$
$F(8) = F(6) - 4 = 7 - 4 = 3$
$F(6) = F(4) - 3 = 10 - 3 = 7$
$F(4) = 4 + F(3) = 4 + 6 = 10$
$F(3) = 3 + F(2) = 3 + 3 = 6$
$F(2) = 4 - 1 = 3$
$F(-2) = 4 - 1 = 3$

So $F(F(F(F(10)))) =$ 7
$F(F(F(-2))) =$
$F(F(3)) =$
$F(6) = 7$

10. 05-06 C1 Recursive Functions

The sequence in length times the number of segments is as follows: 96
$16(1) + 8(2) + 4(4) + 2(8) + 1(16) + 0.5(32) = 96$.

11. 06-07 C1 Recursive Functions

In the first step $3^0 = 1$ triangle is removed. 81
In the second step $3^1 = 3$ triangles are removed.
In the third step $3^2 = 9$ triangles are removed.
In this manner in the fifth step $3^4 = 81$ triangles are removed.

12. 06-07 C1 Recursive Functions

$f(100,3) = f([\frac{100}{3}], 4) - 3 = f(33,4) - 3 = 0 - 3 = -3$ -3

$f(33,4) = f([\frac{33}{4}],5) - 3 = f(8,5) - 3 = 3 - 3 = 0$

$f(8,5) = f([\frac{8}{5}], 6) - 3 = f(1,6) - 3 = 6 - 3 = 3$

$f(1,6) = (1*6) * (\text{gcf}\,(1,6)) = 6 * 1 = 6$

13. 07-08 C1 Recursive Functions

$f(12,66) = f(132,2) - 12 = 4279$ 4279
$f(132,2) = f(2,66) + 2 = 4291$
$f(2,66) = f(66,1) - 2 = 4289$
$f(66,1) = 66^2 - 1*66 + 1^2 = 4291$

14. 07-08 C1 Recursive Functions

$f(1) = 2$ 1535
$f(2) = 2* f(1) + 1 = 2*2 + 1 = 5$
$f(3) = 2* f(2) + 1 = 2*5 + 1 = 11$
$f(4) = 2* f(3) + 1 = 2*11 + 1 = 23$
…
$f(10) = 2* f(9) + 1 = 2*767 + 1 = 1535$

15. 08-09 C1 Recursive Functions

$f(6,3) = 6 + f(5,3) = 6 + 16 = 22$ 22
$f(5,3) = 5 + f(4,3) = 5 + 11 = 16$
$f(4,3) = 4 + f(3,3) = 4 + 7 = 11$
$f(3,3) = 3 + f(1,3) = 3 + 4 = 7$
$f(1,3) = 1 + 3 = 4$

16. 08-09 C1 Recursive Functions

$f(16) = f(14) - 1 = 12 - 1 = 11$ 13
$f(14) = f(12) - 1 = 13 - 1 = 12$
$f(12) = f(10) - 1 = 14 - 1 = 13$
$f(10) = f(13) + 1 = 13 + 1 = 14$
$f(13) = f(11) - 1 = 14 - 1 = 13$
$f(11) = f(9) - 1 = 15 - 1 = 14$
$f(9) = 2*9 - 3 = 15$
$f(f(f(11))) = f(f(14)) = f(12) = 13$

17. 09-10 C1 Recursive Functions

$f(1) = 1$ $f(2) = 2$ $f(3) = 2^1/2 = 1$ $f(4) = 2^2/1 = 2^2$ $2^{-4.5}$
$f(5) = 2^1/2^2 = 2^{-1}$ $f(6) = 2^4/2^{-1} = 2^5$ $\therefore f(7) = 2^{1/2}/2^5 = 2^{-4.5}$

18. 09-10 C1 Recursive Functions

$f(7) = f(11) - 7 = 19 - 7 = 12$
$f(11) = f(8) + 11 = 8 + 11 = 19$
$f(8) = f(12) - 8 = 16 - 8 = 8$
$f(12) = f(9) + 12 = 4 + 12 = 16$
$f(9) = f(13) - 9 = 13 - 9 = 4$
$f(13) = f(10) + 13 = 0 + 13$
$f(10) = 0$

So $f(f(f(7)))$
$= f(f(12))$
$= f(16)$
$= f(13) + 16$
$= 13 + 16 = 29$

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19. 1**0-11 C1 Recursive Functions**

$f(16) = 2 * f(14) + 3 = -27$
$f(14) = 2 * f(12) + 3 = -15$
$f(12) = 2 * f(10) + 3 = -9$
$f(10) = f(5) - 1 = -6$
$f(5) = f(2) - 1 = -5$
$f(2) = f(1) - 1 = -4$
$f(1) = 1 - 4 = -3$

-27

20. 1**0-11 C1 Recursive Functions**

$f(20) = f(14) - 4 = 8 - 4 = 4$
$f(14) = f(8) - 4 = 12 - 4 = 8$
$f(8) = f(f(11)) + 1 = f(11) + 1 = 11 + 1 = 12$
$f(11) = f(5) - 4 = 15 - 4 = 11$
$f(5) = 3 * 5 = 15$

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21. 1**1-12 C1 Recursive Functions**

$f(16) = f(f(13)) + 4 = f(0) + 4 = -5 + 4 = -1$
$f(13) = f(f(10)) + 4 = f(1) + 4 = -4 + 4 = 0$
$f(10) = f(f(7)) + 4 = f(2) + 4 = -3 + 4 = 1$
$f(7) = 7 - 5 = 2$ $\quad f(2) = 2 - 5 = -3$
$f(1) = 1 - 5 = -4$ $\quad f(0) = 0 - 5 = -5$

-1

22. 11-12 C1 Recursive Functions

$f(18, 6) = f(15, 7) - 2 = -5 - 2 = -7$ -7

$f(15, 7) = f(12, 8) - 2 = -3 - 2 = -5$

$f(12, 8) = f(9, 9) - 2 = -1 - 2 = -3$

$f(9, 9) = f(10, 9) = -1$

$f(10, 9) = f(7, 10) - 2 = 1 - 2 = -1$

$f(7, 10) = 3*7 - 2*10 = 21 - 20 = 1$

23. 12-13 C1 Recursive Functions

$f(20) = f(10) - 1 = 0 - 1 = -1$ So $f(f(f(f(20))))$ -2

$f(10) = f(5) - 1 = 1 - 1 = 0$ $= f(f(f(-1)))$

$f(5) = f(0) - 2 = 3 - 2 = 1$ $= f(f(4))$

$f(0) = 0 + 3 = 3$ $= f(15) = f(10) - 2$

$f(-1) = 1 + 3 = 4$ $= 0 - 2 = -2$

$f(4) = f(2) - 1 = 16 - 1 = 15$

$f(2) = f(1) - 1 = 17 - 1 = 16$

$f(1) = f(-4) - 2 = 19 - 2 = 17$ $f(-4) = 16 + 3 = 19$

24. 12-13 C1 Recursive Functions

$f(2,12) = f(f(6,1), f(10,2)) + 2 - 2 = f(7,12) + 2 = 31 + 2 = 33$ 33

$f(6,1) = 7$ $f(10,2) = 12$

$f(7,12) = f(f(11, 1), f(10,2)) + 7 = f(12,12) + 7 = 24 + 7 = 31$

$f(11,1) = 12$ $f(10,2) = 12$

$f(12,12) = f(24,12) - 12 = 36 - 12 = 24$

$f(24,12) = 24 + 12 = 36$

25. 13-14 C1 Recursive Functions

$f(16) = f(13) - f(15) = 0 - 28 = -28$ -28

$f(13) = f(10) - f(12) = 13 - 13 = 0$

$f(15) = f(12) - f(14) = 13 - (-15) = 2$

$f(10) = f(7) - f(9) = 11 - (-2) = 13$

$f(12) = f(9) - f(11) = -2 - (-15) = 13$

$f(14) = f(11) - f(13) = -15 - 0 = -15$

$f(9) = f(6) - f(8) = -4 - (-2) = -2$

$f(11) = f(8) - f(10) = -2 - 13 = -15$

$f(7) = f(4) - f(6) = 7 - (-4) = 11$

$f(6) = f(3) - f(5) = 5 \quad 9 = -4$

$f(8) = f(5) - f(7) = 9 - 11 = -2$

$f(3) = 2*3-1 = 5$ $f(4) = 2*4-1 = 7$ $f(5) = 2*5-1 = 9$

26. 13-14 C1 Recursive Functions

$f(13) = f(11) + 1 = 12+1 = 13$ So $f(f(f(13)))$ 13
$f(11) = f(9) + 1 = 11+1 = 12$ $= f(f(13))$
$f(9) = f(7) + 1 = 10+1 = 11$ $= f(13)$
$f(7) = 2* f(6) – 2 = 2*6-2=10$ $= 13$
$f(6) = 2* f(5) – 2 = 2*4-2=6$
$f(5) = 2* f(4) - 2 = 2*3-2=4$
$f(4) = 4 – 1 = 3$

27. 14-15 C1 Recursive Functions

$f(15) = 2 * f(12) – 4 = 2*18 – 4 = 32$ 32
$f(12) = 2 * f(9) – 4 = 2 * 11 – 4 = 18$
$f(9) = f(10) + f(7) = 6 + 5 = 11$
$f(10) = 2 * f(7) – 4 = 2 * 5 – 4 = 6$
$f(7) = f(3) – 1 = 6 – 1 = 5$
$f(3) = 3 + 3 = 6$

28. 14-15 C1 Recursive Functions

$f(15,12) = f(11,13) + 4 = 8 + 4 = 12$ 12
$f(11,13) = f(12,9) + 4 = 4 + 4 = 8$
$f(12,9) = f(8,10) + 4 = 0 + 4 = 4$
$f(8,10) = f(11,7) + 2 = -2 + 2 = 0$
$f(11,7) = f(6,9) + 4 = -6 + 4 = -2$
$f(6,9) = f(9,6) + 2 = -8 + 2 = -6$
$f(9,6) = f(12,3) + 2 = -10 + 2 = -8$
$f(12,3) = f(2,10) + 4 = -14 + 4 = -10$
$f(2,10) = 3*2 −2*10 = 6 – 20 = -14$

29. 15-16 C1 Recursive Functions

Original size: 27 x 27 = 729 64
First subdivision: colored squares: (9 x 9) x 5 squares = 405
Second subdivision: colored squares (3 x 3) x 20 squares = 180
Third subdivision: colored squares (1 x 1) x 80 squares = 80
So 729 – (405 + 180 + 80) = 729 – 665 = 64 uncolored squares.

30. 15-16 C1 Recursive Functions

$f(16) = f(6) + 3 = 15 + 3 = 18$
$f(6) = f(5) + 2 = 13 + 2 = 15$
$f(5) = f(4) + 2 = 11 + 2 = 13$
$f(4) = f(3) + 2 = 9 + 2 = 11$
$f(3) = f(2) + 2 = 7 + 2 = 9$
$f(2) = f(1) + 2 = 5 + 2 = 7$
$f(1) = 1 + 4 = 5$
$f(18) = f(7) + 3 = 17 + 3 = 20$
$f(7) = f(6) + 2 = 15 + 2 = 17$
$f(20) = f(8) + 3 = 10 + 3 = 13$
$f(8) = f(2) + 3 = 7 + 3 = 10$
$f(13) = f(4) + 3 = 11 + 3 = 14$

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So $f(f(f(f(16))))$
$= f(f(f(18)))$
$= f(f(20))$
$= f(13)$
$= 14$